
UNIT 4 PRODUCTIVITY

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4.1 INTRODUCTION

Production and productivity are two most common and popular terms used in manufacturing sector. This is because that well being of a nation and standard of common people are associated with the effective utilisation of production and improvement in the productivity. This unit is intended to focus on the basic concepts of productivity and other areas of productivity engineering and production management. One should not confuse production with productivity.

Productivity is one of the most commonly used term in industrial engineering. It is a measure of how well resources are utilised to produce output. The term productivity symbolises the following :

- It relates output to inputs in any conversion system, where some value addition is performed on the input resources.
- It is a quantitative measure of performance.
- It integrates performance aspects of quality, efficiency and effectiveness.

Productivity improvement is not just doing things better; more important is to do the right things better. This unit aims to identify the major factors or right things, which should be the main concern of management desiring of productivity improvement. Productivity improvement depends upon how successfully we identify the use of the major factors of the socio-production system.

There are number of factors which affect the productivity of a system/country and not a single factor is independent of others. The importance is to be given to the productivity related to each of the resources – land, material, machine or men depends on the enterprise, the industry and possibly the country. For example, the industries where labour costs are low as compared to material cost, the capital invested in plant and

equipment and better use of materials or plant may give the greatest scope for productivity improvement than putting emphasis on labour productivity. Therefore, for making improvement in productivity, we must know the factors which affect the productivity of a system/country.

Productivity is the optimized utilization of all available resources, investigation into the best known resources and generating new resources, through creative thinking, research and development and by the use of all possible improvement techniques, methods and approaches for the production and distribution of quality goods and services. To improve productivity, it is essential to measure it. Productivity measurement is planned and systematic activity that is used to assess and quantify the status of productivity at a specified period of time. The measurement will help :

- (a) to know the present state of productivity,
- (b) to initiate the action plan needed to improve productivity, and
- (c) to check the effectiveness of action plan by measuring the changed state of productivity.

In this unit, you will study the various definition related to productivity measurement, and the techniques of productivity measurement. It must be emphasized here that the measurement systems in themselves have no inherent value. Their value is realised only when they are incorporated into management operations on systems.

Objectives

After studying this unit, you should be able to

- describe the concept of productivity and production,
- calculate productivity index for an organisation,
- know the ways to improve the productivity,
- explain factors affecting the productivity,
- describe the relationship of productivity with quality, employment, technology and living standards, and
- appreciate the need of productivity measurement.

4.2 FUNDAMENTALS OF PRODUCTION

Production is a conversion process that converts inputs (raw materials, man, equipment, tools, energy, money, etc.) into finished products that have value for the customers and consumers. The conversion process usually involves a series of steps, each step adding value to the inputs and thereby bringing the material close to the desired state of product.

There is a wide variety of products required by the market place. These products are manufactured in different types of industries using different technologies, raw materials, etc. The large list of industrial units can be categorized based upon :

- (a) Nature of products
- (b) Nature of end use of product
- (c) Nature of production system

Classifying the industrial units help in underlying common features, of designing manufacturing strategies, production planning and production control procedures and systems. The efficiency and effectiveness of any production system lies in the use minimum utilisation of resources to produce the desired products of desired quality of desired time. This leads us to the definition of productivity, which is defined as the ratio at output to input. Putting in simple words, improving productivity means

- (a) producing the same output with less inputs,
- (b) producing more output with same inputs,
- (c) producing more output with less inputs or
- (d) producing substantially high output with marginal increase in inputs.

4.2.1 Classification of Industrial Units Based on Nature of Products

A list of basic industries that produce goods is given in Table 4.1. The organisations represented in this table can be divided into two types, depending on the nature of their production operations. The two types are the manufacturing industries and process industries. *Manufacturing industries* are typically identified with discrete item production : machine tools, bicycles, cars, components that go into these products. The *process industries* are represented by petroleum products, food processing, steel, soap, textiles, cement, chemicals and plastics.

Table 4.1 : List of Basic Industries

Manufacturing Industry	Process Industry
Automotives	Cement
Machine tools	Chemicals
Aerospace	Fabric
Ships and Rail coaches	Metals : Aluminium, Copper, Steel
Medicines	Textile
Soaps and Cosmetics	Paper

Classification of Industrial Units Based on the End Use of Product

There are three ways to classify companies on the basis of the end use of the products.

These three types form a connecting chain in the transformation of natural resources and basic raw materials into products for the consuming public.

Basic Producer

They take the natural resources and transform these into raw materials used by other industrial manufacturing units as input. For example, steel producers transform iron ore into steel ingots.

Converter

They are the intermediate link in the chain of transformation. The converters take the output of the basic producers and transform these raw materials into various industrial products and some consumer items. For example, steel ingots are converted into bar stock or sheet metal. Chemical firms transform petroleum products into plastics for moulding. Paper mills convert wood pulp into paper. The output of the converter firms is not complicated, they are simple products.

Fabricator

The fabricators provide the goods to be used by the consumers. They are the last stage in the conversion process. These firms fabricate and assemble final products. The metal bars are transformed into automobile parts and components. Automobile body panels are manufactured from metal sheets. Engine block is made from metal stock. In foundry and casting, the metal is melted and then converted into desired shapes for machining and used in assembly. By using moulding machines, the plastics are molded into various shapes. By this classification, fabricators include both the firms that produce the component and those which assemble component into consumer goods.

Classification of Industrial Units Based on Nature of Production System

In other classification organisations are classified according to the quantity and variety of product made.

Mass Production System

This is a continuous, dedicated manufacturing of identical products. This system is characterised by very high production rate and a limited variety. In this system, the equipment is special purpose rather than general purpose.

Batch Production System

This involves the manufacturing of medium sized lots of the same item or products. The lots are produced in economic order quantity at regular intervals. The plant operates to build up an inventory of the item, then it changes over to other items. When the stock of first item is depleted, production is repeated to buildup the inventory again. The manufacturing equipment used in batch production in general purpose but designed to handle several operations for high production rate. Flexible manufacturing systems are best suitable for batch production.

In Job Shop Production System

The lot size is very small and the variety is very large. The production equipment consists of general purpose machines and require skilled manpower to operate these equipment for variety of jobs.

The productivity, in general, of mass production system is high as compared to batch production system and job shop production system. However, the mass production system is inflexible which makes it unsuitable for present day competitive and customer oriented manufacturing environment. With the application of computer, for the last four decades, efforts are being made to improve the flexibility of mass production system and to improve the productivity of Batch Production System and Job Shop Production System. Following are the various concepts being tried and implemented :

SAQ 1

What are the different basis of classifying industrial/production units?

4.3 DEFINITION OF PRODUCTIVITY

Productivity as defined by Organisation of European Economy Cooperation (OEEC) is :

Productivity is the quotient (ratio) obtained by dividing output by one of the factors of production. Thus, it is possible to speak productivity of capital, labour, space, energy and raw materials according to whether output is being considered in relation to capital, labour, space, energy or raw materials respectively.

It is important to note that productivity is often a misunderstood term. Any specific value of productivity carries relevant meaning only when it is known that the productivity has been calculated in what respect. For example, the productivity of labour and productivity of capital should not be directly compared. You will be explained this point later in this unit.

Historically productivity measures relate to the physical and technical aspects of a productive process. A formal organisation was created to handle problems of specialisation, departmentation and technical co-ordination of jobs in an organisation.

4.3.1 Causes of Low Productivity

For evolving specific measures for improving productivity, it will help the management to analyse on continuing basis, the causes of low productivity and at the same time to keep abreast with the latest management and productivity techniques.

In the Indian context, the causes of low productivity have their origin in two distinct sources. The first category consists of the exogenous or external factors like the shortages of essential inputs-power, raw materials, transport facilities, etc. – over which the management of an enterprise has little or no control. The second basket contains the endogenous or internal factors mainly in the form of system deficiencies preventing the optimum utilisation of resources.

Since the management has very little control on external factors and their adverse effect on productivity, the challenge before the Indian managers lies in overcoming the internal causes of low productivity.

4.3.2 Utility of Increasing Productivity

Constant endeavours should be made to increase the productivity. Higher productivity has manifold advantages.

To the Individual Concern

- (a) Higher productivity means more wages directly to the piece-workers and more production bonus to all workers. It means satisfied staff and harmonious staff relations.
- (b) The factory earns more profit because of the reduction in costs.
- (c) Continuous higher productivity may induce the management to reduce selling prices so that sales and production may increase.

To the Industry

Higher productivity in some concerns will enable less efficient firms to follow them for their own survival.

To the Government

- (a) Higher profits earned by factories will bring more revenue to the government by taxation.
- (b) Export trades may develop bringing more foreign exchange to the nation.
- (c) Overall higher productivity will raise an all round living standard of labour.

SAQ 2

- (a) What is productivity?
- (b) What are the different types of productivity?
- (c) What are the reasons of low productivity?
- (d) Write the basic features of all the three approaches to productivity.
- (e) Identify the internal as well as external factors of low productivity for an organisation?

4.4 PRODUCTIVITY MANAGEMENT

Productivity management at the organisational level is a strategic issue with long term implications. It should be institutionalised as an on-going and self sustaining process for the prosperity of the organisation. A well-known organisation framework for productivity consists of the following elements :

Goals and Objectives

The goals and objectives of the organisation if strategically focussed and well communicated can provide clarity in deciding the operating plans. They provide cost reduction measures, productivity improvement targets and improve the organisational processes.

Rewards and Recognition

Rewards related to productivity enhancement and recognition of successful individuals and teams encourage the employees and positively reinforced their behaviour and actions.

Information

Information and the analysis of information helps to assess the division's progress relative to operating plans and to identify the specific areas for improvement.

Employees Involvement

Employees involvement in small group activities and in cross-functional teams provide opportunities to bring about improvement and influence on management decisions. Management attitude towards the well being of employees, critical information sharing practices, open communication channels and the organisational climate as a whole are vital to success.

Assets

The assets of an organisation include employees, materials, equipment and facilities. To improve the effectiveness of the organisation's assets the typical issues include basic education, skill training, cost of quality, equipment capacity utilization and convert operating practices.

Leadership

Leadership occupies the hub of the productivity wheel, since it drives the other elements of the improvement cycle.

SAQ 3

What are the various elements of productivity improvement?

4.5 DIFFERENCE BETWEEN PRODUCTIVITY AND PRODUCTION

Production and productivity are different terms and carries different meanings. It is not correct to assume that higher production will always lead to higher productivity or vice-versa.

Production is related to the efficient utilisation of input resources into produced output in the form of value added goods or services. You can also say that production is related to the activity of producing goods or services. It is a process (or system) of converting input into some useful, value added output. Production is therefore a measure of output

produced. It does not emphasise on how well the input-resources are utilized, whereas Productivity is, ratio of output produced to the input used. It focuses on how well the input resources is used for conversion of input into output. Following examples will illustrate you the difference between production and productivity.

Example 4.1

Compute the productivity per machine hour with the following data. Also draw your interpretation.

Month	No. of Machines Employed	Working Hours	Machine Hours	Production Unit
March	400	225	90,000	99,000
April	500	200	100,000	100,000
May	600	250	150,000	135,000

Solution

We know P = Productivity per machine hour,

$$= \frac{\text{Production units}}{\text{Machine hours}}$$

$$\text{For March } p = \frac{99,000}{90,000} = 1.1$$

$$\text{For April } p = \frac{100,000}{100,000} = 1$$

$$\text{For May } p = \frac{135,000}{150,000} = 0.9$$

Interpretation : Though the total production units are increasing, the productivity is declining.

Example 4.2

A company is manufacturing 24,000 components per month by employing 100 workers in daily 8 hour shift. The company gets additional order to produce additional 6000 components. The management decides to employ additional workers. What will be production and productivity level when the number of additional employed workers are (i) 30, (ii) 25, and (iii) 20.

Solution

Present production = 24,000 components

$$\begin{aligned} \text{Present productivity (of labour)} &= \frac{\text{Present Production (i.e. output)}}{\text{Total man hours (i.e. input)}} \\ &= \frac{24,000 \text{ components}}{(100 \text{ workers}) (8 \text{ hours}) (30 \text{ days of the month})} \\ &= \frac{24,000}{24,000} = 1 \text{ component/man hour} \end{aligned}$$

With increased order

(a) When additional 30 workers are hired

Production = 24,000 + 6000 = 30,000 components

$$\text{Productivity (of labour)} = \frac{\text{Increased total production}}{\text{Total man hours}}$$

$$= \frac{30,000}{(100 + 30) (8) (30)} = 0.96 \text{ component/man hour}$$

- (b) When additional 25 workers are hired

$$\text{Production} = 24,000 + 6000 = 30,000 \text{ components}$$

$$\text{Productivity (of labour)} = \frac{30,000}{(100+20)8 \times 30} = 1 \text{ component/man hour}$$

- (c) When additional 20 workers are hired

$$\text{Production} = 24,000 + 6000 = 30,000 \text{ components}$$

$$\text{Productivity (of labour)} = \frac{30,000}{(100+20)8 \times 30}$$

$$= \frac{30,000}{120 \times 8 \times 30} = 1.04 \text{ component/man hour}$$

In this example, it is clear that production has increased 6000 units. Therefore,

$$\text{increase in production} = \frac{30,000 - 24,000}{24,000} \times 100 = 25\%$$

In the above case you must have observed that how the productivity is varying with the change of input labour while production target is kept constant. It should also be clear from the example that the labour productivity falls below the initial level of 1 component per man-hour if more than 25 workers are hired. You may treat this level of additional manpower as break-even level from the point of view of labour productivity. Therefore, for constant production and working hours, you may conclude that not more than 25 workers should be hired for meeting this increase production requirement.

We have tried to make clear following three points to you from the above example :

- (a) Production and productivity are two different terms.
- (b) Increase in production does not necessarily mean increase in productivity.
- (c) Productivity is always associated with the context in which it is calculated.

For example, in the above case we have calculated labour productivity. In another case, some one may like to know about material productivity or energy productivity. If labour productivity has decreased due to some reason, it may not be always meant that other forms of productivity for example material productivity will also fall. Various forms of productivity will be explained in the coming sections of this unit.

SAQ 4

- (a) What is the difference between production and productivity?
- (b) Why increase in production always does not lead to increase in productivity of a production unit?

4.6 SCOPE OF PRODUCTIVITY

You must have read that India slipped from its 138th position to 139 during 1998 when study was undertaken for 174 countries. Another reflection of this dismal picture is that 86% of the total global consumption is enjoyed by 20% of the world's population. All these had been possible by the developed countries due to special attention paid to integrate the productivity dimensions with national and enterprise level policy formulation and implementation.

The thrust of productivity movement is no longer confined to shop floor activities in the industry, and extends even to rural and community developments, and has new focus areas like women and informal sector, and local communities. Another significant dimension of productivity is sustainable development, covering environmental issues, energy conservation, pollution prevention, cleaner production techniques and the concept of green productivity and green services, denoting environment friendly process, inputs and products.

Experience has shown that productivity of social system and services sector not only affect the productivity of industrial sector but also quality of life in a significant manner. Environment enhancement, unemployment, poverty eradication, social justice, social welfare policies, education system, health system are some areas that engage our attention.

4.7 EFFECT OF PRODUCTIVITY ON SOCIETY

It would be worthwhile to examine the gains from the productivity improvement to various sections of the society. If productivity is increased, the distributive efficiency of the economy is maintained and the gains are properly shared, productivity can benefit all sections of the society. This is why productivity is a subject in which employers, labour, community and the government all have strong mutual interests.

Productivity improvement results in direct increase in the standard of living under conditions of distribution of productivity gains according to contribution. At present, it would not be wrong to state productivity is the only important worldwide source of real economic growth, social progress and improved standard of living.

Productivity as a subject involves the employers, labour, community and the government. These gains could be shown as follows :

Employer's Gains

Since productivity reduces cost and increases competitive capacity of the product, employer's profitability is best ensured through expansion of industry or setting up new industry.

Labour's Gains

Increased wages are possible to any substantial extent only through higher productivity. In India initial productivity levels are already very low and therefore there are tremendous scope to increase the earnings of workers through promotion of productivity in all enterprises or industry.

Community Gains

Increased productivity resulting in more production with reduced cost and better quality means not only increased wealth but greater satisfaction to the community due to adequate supply of goods and services at reasonable rates. Further, higher productivity resulting in proper utilisation of resources will create surplus resources which could be invested for industrial and agricultural development, thereby generating more employment opportunities. Infact higher productivity alone can solve our problems like inflation, recession, unemployment and poverty.

With increased production the income of the government will also increase in the form of more tax collections which could be spent on promotions of services like education, public health, public transport, etc. Efforts in the direction of increasing production in the existing public utility services itself will improve efficiency and create goodwill in the public, thereby ensuring political and social stability in the society.

4.8 FACTORS AFFECTING PRODUCTIVITY

Productivity improvement depends upon how successfully we identify and use the main factors of the socio-production system. The productivity factors can be classified as external factors and internal factors. In general, external factors are non-controllable and internal factors are within the control of management. This classification will help managers to distinguish those factors which they can control easily. Thus, the two major categories of productivity factors are :

- (a) External factors (non-controllable), and
- (b) Internal factors (Controllable).

To deal with all these factors, we require different institutions, people, techniques and methods. For example, any performance improvement drive, which plans to deal with external factors affecting management of the enterprise, it would be better if such factors are included at the planning phase of the programme and if management is convinced along with other affected parties.

Factors which are external and non-controllable for one enterprise are often internal to another. Factors external to an enterprise, for example, could be internal to government, national or regional institutions, associations and pressure groups. Government can improve tax policy, develop better labour legislation, provide better access to natural resources, improve social infrastructure, price policy and so on, but individual organisations cannot. Factors external to an enterprise are of interest to that enterprise because an understanding of them can motivate certain actions which might change an enterprises behaviour and is productivity in the long run. Figure 4.1 gives the integrated scheme of factors constituting a major source of productivity improvement.

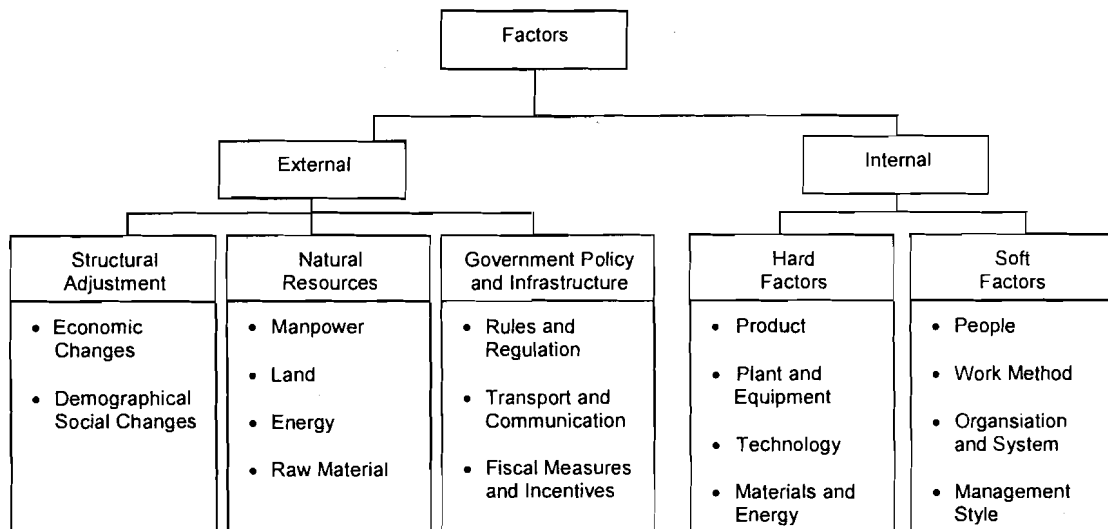


Figure 4.1 : Scheme of Factors Affecting Productivity

4.8.1 Internal Factors

Since some internal factors are more easily changed than others, it is useful to classify them into two groups : hard factors (not easily changed) and soft factors (easily changed). The hard factors include product, technology, equipment and raw materials,

while the soft factors include the labour force, organisational systems and procedures, management styles and work methods. This classification helps to build priorities, which factors can easily be dealt with and which factors require stronger financial and organisational interventions. A brief description of some key aspects of the internal factors is presented below.

Hard Factors

Product

The extent to which the product meets customers requirement is meant by product factor productivity. 'Use value' is the amount that the customer is prepared to pay for a product of given quality. 'Use value' can be improved by better design and specification. Many companies around the world fight a constant battle to incorporate technical excellence into marketable products. They continually redesign products which are in the market. Product 'place value', 'time value' and 'price value' refer to the availability of the product at the right place at the right time and at a reasonable price.

Research and development (R&D) is a vital contribution to improve product design. Studies of R&D and its effect on productivity show an additional relationship. Research can turn up entirely new principle that permit a function to be performed in a new way at a much lower cost. For example, the today's office copy machines made the copying of documents so easy for secretaries without the use of carbon paper or the need of retyping.

Simplification and standardisation of the product are other design factors that make possible higher productivity in the factory. This helps in reducing engineering cost, tooling cost, provides economics of scales and an effective after sales service. Value engineering helps in meeting the desired functions by using alternative design and materials of manufacturing. For example, the size of today's 500,000 kVA transformer has same dimensions and weight as that of 72,000 kVA transformer forty years ago. This was made possible by continuous improvement using value engineering technique.

Plant and Equipment

Once the product is designed, then next aspect which needs attention for productivity improvement is related to plant and equipment. The equipment used – machines, tools, conveyors and robots, etc. – play a key role in a productivity improvement programme through :

- good maintenance,
- operating and plant and equipment in optimum process conditions,
- increasing plant capacity by eliminating bottlenecks and by corrective measures, and
- reducing idle time and making more effective use of available machines and plant capacities.

Plant and equipment productivity can be improved by paying attention to optimization, age, modernisation, cost, investment, internally produced equipment, capacity maintenance and expansion, inventory and control and production planning and control.

Technology

Technological innovation constitutes an important source of achieving higher productivity. Increased volume of goods and services, quality improvement and new marketing methods, etc. can be achieved through increased automation and information technology. Automation can also improve material handling, storage, communication systems and quality control.

During the last two decades, considerable productivity improvement has been realised through the use of automation and current development in information technology. Significant examples of the applications of this technology are the development of automatic down time recording systems and automatic lubricating systems which have reduced the idle time of man machines. New technology is normally introduced as a result of such productivity improvement programmes as fighting obsolescence, process design, R&D and the training of scientists and engineers.

The importance of machinery and equipment on productivity in service sector can be demonstrated by looking at a kitchen or a laundry room. A generation ago, a home maker had to devote a full day in washing the cloths and she was not able to perform other activities. Now, with automatic washing machines, other activities can also be carried out along with washing of the clothes. Productivity improvement has eradicated the term 'washing day' right out of our vocabulary.

Materials and Energy

Even small efforts to reduce materials and energy consumption can bring remarkable results. These vital sources of productivity improvement include raw materials and indirect materials (process chemicals, lubricants, fuels, spare parts, engineering materials and packing materials, etc.). Improvement aspects of material productivity include :

- material yield : output of useful product or service per unit of material used,
- use and control of wastages of scraps,
- upgrading of materials by initial processing to improve utilisation in the main process,
- use of lower grade and cheaper materials,
- import substitution,
- improving inventory turnover ratio to release funds tied up in inventories for more productive uses, and
- developing alternate and reliable sources of supply.

Soft Factors

People

As the principal resources and the central factor in productivity improvement drives, the people in an organisation and trade union member. Each role has two aspects : application and effectiveness.

Application is the degree to which people apply themselves to their work. People differ not only in their ability but also in their will to work. The ability to work can be improved by training and education. The will to work can be explained by a law of behaviour : motivation decreases if it is either satisfied or blocked from satisfaction. The following factors can be considered in order to stimulate and maintain motivation :

Value : A set of values conducive to higher productivity should be developed in order to bring about changes in the attitude of managers, engineers and workers.

Motivation

: Motivation is the state of an individual's perspective which represents the strength of his prosperity to exert effort towards some particular behaviour to organisational performance in order to improve productivity. Worker's contributions in increasing productivity should be rewarded immediately, not only in the form of money, but also by recognition, involvement and learning opportunities.

Productivity**Incentive Schemes**

: The effective incentive schemes can be used as a means of significant improvement in productivity. Wage incentives must always be related to the amount of change accomplished.

Worker's Participation :

Productivity can also be improved by eliciting cooperation and participation from workers. Worker's participation in goal setting, for example, has been quite successful in many organisations and countries.

Communication

: Human relations and hence the productivity can be improved by reducing the complexity of communication procedures and by minimising conflicts.

Creativity

: Labour productivity can be tapped only if management encourages workers to apply their creative talents by taking a special interest in their problems and by promoting a favourable social climate.

Ability of Workers

: Effectiveness of productivity efforts depends upon the ability of workers to the extent to which the application of human effort brings the desired results in output and quality. It is a function of method, technique, personal skill, knowledge, attitude and aptitude. This ability to do a productive job can be improved through training and education, job rotation and placements, systematic job progression and career planning.

In any organisation, those who work will tend to get more work. Those who do not work will get pay perquisite and promotions. In general, people in the organization can be divided into seven types (Table 4.2) from work culture point of view. Different strategies have to be prepared to harness good and reform or remove the escapist.

Table 4.2

	Type	Action
a	High fliers	Use them as trainers/leaders
b	Innovators	Give them important/special assignment
c	Hard workers	Encourage them
d	Fence sitter	Communicate about the future programme, to reform them as hard workers
e	Laggards	Train, counsel and help
f	Speed breakers	Isolate them and keep under observation
g	Check posts/hurdles	These are the people with clout. Transfer them to break links

To summarise, the following key approaches, methods and techniques can be used to improve labour productivity : wages and salaries, training and education, rewards and incentive schemes, participation and cooperation, etc.

Work Method

Improved work methods, especially in developing economics where capital is scarce, technology intermediate and labour intensive methods dominant, constitute the most promising area for productivity improvement. Work method techniques aim to make manual work more productive by improving the ways in which the work is done, the human movements are performed, the tools are used, the workplace is laid out, the material is handled and the machines are employed. Work methods are improved by systematically analysing the present methods, eliminating unnecessary work and performing the necessary work more effectively with less efforts, time and cost.

Organisation and Systems

The well known principles of good organisation such as unity of command, delegation and span of control are intended to provide for specialisation and intended division of work and coordination within the enterprise. The organisation needs are to be dynamically operated and focused towards objectives and they must be maintained, serviced and reorganised from time to time to meet new objectives.

Rigidity is one reason for the low productivity in many organisations. Due to rigidity, organisations fail to anticipate and respond to marked changes. They ignore new capacities in the labour force, new developments in technology and other external changes. Rigid organisations lack good horizontal communication. This slows down decision making process and inhibits delegation of authority close to the point of action, encourages inefficiency and bureaucracy.

Management of organisation by functions and air tight departments also inhibits changes. For example, the decision making steps may have been designed for a particular existing technology, for a definite product or service mix. Things have now changed, but procedures have been survived because managers want to minimise changes. Dynamism and flexibility should be incorporated into the system design in order to maximise productivity. No system, however, well designed, is efficient in all situations.

Management Style

It is observed that 75 percent of productivity gains, in most of the organisations, are possible by effective use of resources under enterprise control by the management. Productivity experts believe that as much as 85 percent that common problems are related to the quality and productivity problems that can be resolved by the management, not by the individual worker. As there is no perfect management style, effectiveness depends upon when, where, how and to whom a manager applies a style. Management styles and practices influence organisational design, personnel policies, capital cost, source of capital, budgeting systems and cost control techniques.

4.8.2 External Factors

External factors include government policies and institutional mechanism, political, social and economic conditions, the business climate, the availability of power, workers, transport, finance, communication and raw materials. They affect individual enterprise

productivity, but the organization concerned cannot actively control them. These factors should be understood and taken into considerations by management while planning and implementing productivity programmes.

Considering all the social, political, economic and organisational links among consumers, workers, managers, government and different pressure groups and between institutions and organisational infrastructure, it is useful to discuss here the main macro-productivity factors which speed or hinder productivity improvement processes. As productivity largely determines real income, inflation, competitiveness and people's well being, policy makers try hard to discover the real reasons for productivity growth and decline. The three main groups of macro-productivity factors are as follows :

- (a) Structural Adjustment
 - (i) Economic changes
 - (ii) Demographic and social changes
- (b) Natural Resources
 - (i) Manpower
 - (ii) Land
 - (iii) Energy
 - (iv) Raw materials
- (c) Government and Infrastructure
 - (i) Institutional mechanisms
 - (ii) Policies and strategies
 - (iii) Infrastructure
 - (iv) Public enterprises

Structural Adjustment

Economic and demographic changes in the society often influence national and enterprise productivity independently of enterprise management. In the long term, however, this interaction is two-way. Just as structural changes affect productivity, productivity changes also modify structure. Such changes are not only the result, but also a cause, of economic and social development. Understanding these changes helps to improve government policies, makes enterprise planning more realistic and purpose oriented and helps to develop the economic and social infrastructure.

Economic Changes

The most important economic changes are in employment patterns and in the composition of capital, technology, scale and competitiveness.

- Shift of employment from agriculture to manufacturing industry has caused an economy-wide increase in productivity. In many developing countries, these shifts will continue to be a source of high growth in productivity in future, since more people will move from the low productivity agricultural sector into manufacturing sector.
- The management of research, development and technology and the implementation of new methods, techniques, products and processes can significantly influence productivity. Introduction of new technology often requires foreign investment.

Demographical Social Changes

Productivity and wages in the developing countries tend to be lower and the total cost of production is competitive. Two different and somewhat contradictory pressures influence productivity. On the one hand, producers

in developing countries must try to increase productivity to hold down production costs, on the other hand low wages of workers encourages to use more labour rather than investing heavily in capital and equipment. This tends to reduce the labour productivity.

Increasing percentage of women in labour force is an important social factor. The retirement age may drift upward, as health and longevity improve. Economic pressures may also persuade many older people to stay in the labour force.

Education is another social factor which affects productivity. Cultural values and attitude can promote or hinder productivity. For example, the Chinese are known for their belief in hard work, their entrepreneurial spirit and their attitude for savings. The Japanese are famous for their ability to seek, accept, assimilate and adapt to changing needs and circumstances, for their team spirit and discipline. It is important to study and understand these beliefs, attitudes and traditions, all of which change with new technology and economic developments.

Natural Resources

The most important natural resources are manpower, land, energy and raw materials. A nation's ability to generate, mobilize and use these resources are important to improve productivity. Availability of natural resources is a major factor in facility location decision in any industry which affects productivity of the industry.

Manpower

Manpower is the most important natural resource. Several countries, which lack other natural resources have invested in human resource development for improving workers skills and changing their attitude through training, enhancing their education and motivating them by giving incentives and recognition. Countries with higher per capita GNP generally have a better trained and educated manpower.

Land

For industrial expansion and intensive farming, land is an important natural resource. The rising cost of energy-intensive agricultural inputs, the limited availability of new land and the urgent need for more careful husbandry to prevent corrosion of land argue for more prudent use of available land.

Energy

The supply of energy influences capital and labour productivity. The high cost of energy and cut back on energy use decrease capital investment and encourage to increase the use of labour. Thus, the demand of labour tends to follow energy prices.

Raw Materials

The cost of minerals and other raw materials affects the economic rationale for repair, re-use and recycling. These become more compelling when the cost of raw-material rises.

Government and Infrastructure

Government policies, strategies and programmes greatly affect productivity through :

- practices of government agencies,
- regulations such as price control, subsidies, etc.
- transport and communications,

- power, and
- fiscal measures and incentives like taxes and interest rate, etc.

Therefore, several internal and external factors affect the productivity of an organisation. It is the internal factors that are under the control of enterprise management. However, to design good policies, plans and programmes for productivity improvement, all the external factors should be analysed, understood and must be considered.

SAQ 5

- List the factors affecting the national productivity.
- What can a government do for improving the productivity of the nation?
- How does employees participation help in improving the national productivity?

4.9 PRODUCTIVITY MEASUREMENT

As studied earlier, productivity is a universal and simple concept. Since the time, this word was coined in 1776, as a relationship between output and input, it has been variously defined and interpreted to keep pace with changing social and economic context. Conventional view has always emphasized labour dimension and over a period of time, all inputs including energy, material and capital were brought into the ambit of productivity considerations. The output dimension of productivity, though mentioned as an integral part of any productivity definition has not been properly understood in its entirety. In the competitive and customer driven markets, output dimensions reflect the ability to satisfy the emerging needs of competitiveness and reflect societal consideration. The output dimensions can be defined as 'set of results' highest level of performance, organisational objectives, total tangible output, etc. Productivity depends on both quality and features of products and the efficiency with which they are produced.

Thus productivity has evolved into truly multidimensional and complex concept which needs to be interpreted in the context in which it is to be operationalised. The various dimensions which are relevant today in the present environment can be summarised as follows :

- In an era of customer-oriented business, output dimension has assumed added importance. It must incorporate all dimensions such as quality, cost, delivery, after sales service and flexibility which a customer demands from an organization. Goods produced efficiently, which cannot be used or required by the customers indicate poor productivity.
- Products and processes must be environmentally sound, generating low or no waste and pollutants. The emphasis should be to minimise life cycle cost to society by designing the recyclable products and manufacturing them with no waste and pollutants.
- Qualitative aspects of productivity must be emphasised. These would include satisfaction of employees and customers, behaviour courtesy, timeliness and other similar aspects.
- Relationship between quality and productivity must be properly understood. Quality requires freezing of specification while productivity refers to unfreezing and change in status quo.

- (e) Intangible inputs such as knowledge, information and motivation have assumed added importance which play more critical role in productivity improvement.
- (f) Time is the essence of productivity.

In the ultimate analysis, productivity is emerging as a :

- dynamic concept which is constantly changing to be relevant in the context in which it is being operationalised.
- holistic in nature to include contribution of all those concerned with design, manufacture, use and maintenance of products and focused on organisational performance.
- output oriented to focus on relevance of products and services to customer and social needs.
- complex, as its is influenced by a large number of variables at international, national, organisational and individual levels.

Methodologies and models of productivity measurement are essentially the translation of the various definitions and concepts of productivity. Most of the approaches are centered around production and productivity function. Suggested measure of productivity range from simple ratios of production per man hour or per unit of power consumed or standard hours produced per unit of monetary expenses, to more comprehensive measures as – single input productivity indices, multi-factor productivity measures and total factor productivity indexes. Some variations on the theme consider output as value addition. Centred around these basic models of productivity measurement are various methodologies of quantification of both inputs and output, their conversion to common units of measurement, their blending and combination to create composite outputs and composite inputs and reducing the measures to a base reference period. Various approaches have been also worked out to account for inflation in the costs of inputs and outputs. Another set of approaches, classified as profit-centred approaches consider profit as a vital component of productivity.

SAQ 6

- (a) What is the significance of productivity measurement?
- (b) Write the various methodologies used for productivity measurement.

Types of Productivity Measures

The various types of productivity measures are as follows :

Partial Productivity

Partial productivity is the ratio of output to one class of input. For example, labour productivity (the ratio of output to labour input) is a partial productivity measure

$$\text{Partial Productivity Index} = \frac{\text{Output}}{\text{One Factor of Input}}$$

Some of the frequently measured partial productivity measures are :

Labour Productivity

In this formulation, the resource inputs are aggregated in terms of labour hours. Hence the index is relatively free of changes caused by wage rates and labour mix.

Direct Labour Cost Productivity

In this formulation the resource inputs are aggregated in terms of direct labour costs. This index will reflect the effect of both wage rates and changes in the labour mix. However, "constant rupee" may be used to eliminate this distortion.

Capital Productivity

Several formulations are possible. In one, the resource inputs may be the charges during the period to depreciation; in another, the inputs may be the book value of capital equipment.

Direct Cost Productivity

In this formulation all items of direct cost associated with resources used are aggregated on a monetary value basis. Constant rupee may be used.

Total Cost Productivity

In this formulation all resources costs, including depreciation, are aggregated on a monetary basis. Constant rupee may be used.

Foreign Exchange Productivity

In this formulation the only resources cost considered is the amount of foreign exchange required.

Energy Productivity

In this formulation the only resource considered is the amount of energy consumed in British Thermal Unit (btu) or kW, as may be most convenient.

Raw Materials Productivity

In this formulation the numerators are usually weight of product; the denominators are the weight or value of raw materials consumed.

Hence, it may be seen that many different productivity indices may be constructed. The list given above is not exhaustive.

Advantages of Partial Productivity Measure

- (a) It is a good measure to identify areas where improvements may be done.
- (b) Easy to calculate, as independent of other inputs.
- (c) Easy to understand.
- (d) Management finds it easy to understand and pin-point the logic for its improvement.
- (e) It is easy to benchmark (compare) with other industries. For example, the labour productivity of a public limited company producing steel may be compared with labour productivity of a private sector steel producing company.
- (f) Data may be easily generated for this measure of productivity.

Limitations of Partial Productivity Measure

- (a) It can be misleading, if used out of context.
- (b) It does not contain the overall effect of the system performance.
- (c) Focussed areas of improvements are difficult to identify. Therefore, wrong areas of management context may be identified.

- (d) It gives a myopic view of performance. This means, only limited factors are considered to effect the output of performance.
- (e) It misses holistic (or totality) approach.

Total Factor Productivity

Total factor productivity is the ratio of net output to the sum of associated labour and capital inputs. Net output means total output minus intermediate goods and services purchased. Notice that the denominator of this ratio is made up of only the labour and capital input factors.

$$\text{Total Factors Productivity Index} = \frac{\text{Net Output}}{\text{Total Factor Input}}$$

where, Net Output = Output minus intermediate goods and services, and

Total Factor Input = Labour input plus capital input.

Advantage of Total Factor Productivity

- (a) It is relatively easy to compare data from company records.
- (b) Economist prefer this as it is easy to compare in cross-industry context.

Limitations of Total Factor Productivity

- (a) Many important inputs, such as material, energy, etc. are ignored.
- (b) The net-output does not reflect the efficiency of production system in a proper way.

Total Productivity

Total productivity is the ratio of total output to the sum of all inputs factors. Therefore, a total productivity measure reflects joint impact of all the inputs in producing the output. Its formation includes an inclusive statement of the value of the product or service produced and a summary value for all of its inputs. Money dimensions are typically used for both numerator and denominator to allow diverse products and resources to be expressed in equivalent terms.

$$\begin{aligned}\text{Partial Productivity Index} &= \frac{\text{Total Output}}{\text{All Associated Input}} \\ &= \frac{\text{Products} + \text{Services}}{\text{Labour} + \text{Material} + \text{Energy} + \text{Capital}}\end{aligned}$$

From this basic expression, adaptations can be made to more closely represent the functions of a particular organisation. The intent of customizing the index is to reflect the objectives of the firm. For example, one organisation might believe that purchases of raw material is someone else's productivity efforts and should therefore be excluded from the user's input. Another firm might feel that energy is a small and a constant input and can be ignored in the productivity index. An aluminium producer would, of course, feel differently.

Advantages of Total Productivity

- (a) It considers all the outputs and input factors. Therefore, it is more accurate representation of real economic performance of the organisation.
- (b) It is easy for top management to understand the company performance.
- (c) If it is used along with partial productivity, focussed areas of improvements may be identified.

- (d) Easy for cost accountants to compare performances.
- (e) Easy to do sensitivity analysis.

Limitations of Total Productivity

- (a) Difficult to generate company wide and sector wide data.
- (b) Many non-direct measures of input/output are ignored.

Example 4.3

Out of the many possible total productivity models, let ABC Ltd has the following formula :

$$\text{Total Productivity Index} = \frac{\text{Sales} + \text{Inventory change} + \text{Plant}}{\text{Labour} + \text{Material} + \text{Service} + \text{Depreciation} + \text{Investment}}$$

The input and output factors for year 2001 are individually described and numerically demonstrated below :

Sales

Let the sales in 2001 was Rs. 15 crores. Assuming the base year of 2000 and a change in the general price trend of 15% between 2000 and 2001 would result in a general 2001 deflator of $\frac{100}{115} = 0.869$. Then sales of 2001 would amount to $15 \text{ crores} \times 0.869 = 13.04 \text{ crores}$ in constant 2000 rupees.

Inventory Change

Inventory includes work in process and finished goods. Each category of inventory could have its own deflator factor. Assuming a deflator of 0.869 is applicable for all material categories and the total volume of inventories is equal to 0.4 crores during the year, the amount of inventory change has a constant rupee value of $0.4 \times 0.869 = 0.35 \text{ crores}$.

Plant

Capabilities of an organisation are often increased by the output of its employees in ways that increase the worth of the production system but do not directly increase income through sales. Maintenance, R & D and company produced machinery are the examples. The value contributed by the plant improvement can be valued at the internal customer market value. Let the value of the plant is increased by 0.2 crores constant rupee in 2001.

Labour Wages and Other Benefits

Assume 4 crores as the labour cost for 2001. This cost includes compensations for wages, salaries, overtime, vacation, incentives, medical, bonus, etc.

Materials

Let the consumption of materials in 2001 as 6 crores after applying appropriate materials deflators.

Depreciation

Since depreciation charges are not responsive to current price trends, they are included at their current value and is assumed to be 0.4 crores in 2001.

Interest

If we assume 15% for the annual cost of capital to the organisation, total capital assets of 6 crores and a price deflator of 0.869, the constant investment input for 2001 is $6 \times 0.869 = \text{Rs. } 5.21 \text{ crores}$ at constant 2000 prices.

From these output and input factors, the total productivity index for 2001 is calculated as :

$$\text{Total Productivity Index, 2001} = \frac{13.04 + 0.35 + 0.20}{4.0 + 6.0 + 0.4 + 5.21} = 0.87$$

Let the TPI for the year 2000 was 80%. On comprising the TPI for two years, this productivity measurement has more meaning.

$$\text{Productivity change from 2000 to 2001} = \frac{0.87 - 0.80}{0.80} = 8.75\%$$

SAQ 7

What are the various types of productivity measures?

Example 4.4

The data for output produced and inputs consumed for a particular type of a manufacturing organisation are given below in constant money value. Find out the partial, total-factor and total productivity values.

Output	=	Rs. 2000.00
Labour input	=	Rs. 400.00
Material input	=	Rs. 200.00
Capital input	=	Rs. 500.00
Energy input	=	Rs. 100.00
Other expenses input	=	Rs. 50.00

Solution

Partial productivities

$$\text{Labour productivity} = \frac{\text{Output}}{\text{Labour Input}} = \frac{2000}{400} = 5.00$$

$$\text{Material productivity} = \frac{\text{Output}}{\text{Material Input}} = \frac{2000}{200} = 10.00$$

$$\text{Capital productivity} = \frac{\text{Output}}{\text{Capital Input}} = \frac{2000}{500} = 4.00$$

$$\text{Energy productivity} = \frac{\text{Output}}{\text{Energy Input}} = \frac{2000}{100} = 20.00$$

$$\text{Other expenses productivity} = \frac{\text{Output}}{\text{Other Expenses Input}} = \frac{2000}{50} = 40.00$$

$$\text{Total-factor productivity} = \frac{\text{Net Output}}{(\text{Labour} + \text{Capital}) \text{ Input}}$$

$$\text{Net output} = \text{Total Output} - \text{Materials and services purchased}$$

$$= 2000 - (200 + 400 + 100 + 50) = 250$$

$$\text{Total-factor productivity} = \frac{1250}{400 + 500} = 1.389$$

$$\begin{aligned} \text{Total productivity} &= \frac{\text{Total Output}}{\text{Total Input}} \\ &= \frac{2000}{400 + 200 + 500 + 100 + 50} = 1.6 \end{aligned}$$

Example 4.5

Following table gives comparative study of cost of several items for two years. Compute the percentage change in all the productivity indexes.

Item	Year-1	Year-2
Number of outputs (all of one kind) @ Rs. 50,000 per unit	100	200
Direct labour hours (variable with year)	5000	8000
Direct labour cost (in Rs.)	40000	45000
Capital depreciation (in Rs.)	5000	6000
Capital book value (in Rs.)	20000	25000
Total indirect cost (in Rs.)	40000	46000
Foreign exchange used	\$ 450	\$ 100
Energy used (@ Rs. 4 per watt)	500 kw	1800 kw
Raw material used (@ Rs. 1000 per ton)	10 tons	16 tons
Services of consultant hired (in Rs.)	10000	15000

Calculation of Productivity Index

$$(a) \quad \text{Labour productivity index} = \frac{\frac{200}{8000} \times 100}{\frac{100}{5000}} = 125$$

Labour productivity of year 2 is 125% of year 1.

$$(b) \quad \text{Direct labour productivity} = \frac{\frac{200}{45000} \times 100}{\frac{100}{40000}} = 177.8$$

$$(c) \quad \text{Capital depreciation productivity index} = \frac{\frac{200}{6000} \times 100}{\frac{100}{5000}} = 166.7$$

$$(d) \quad \text{Capital book value productivity index} = \frac{\frac{200}{25000} \times 100}{\frac{100}{20000}} = 160$$

$$(e) \quad \text{Total indirect cost productivity index} = \frac{\frac{200}{46000} \times 100}{\frac{100}{40000}} = 174$$

$$(f) \quad \text{Foreign exchange productivity index} = \frac{\frac{200}{100} \times 100}{\frac{100}{4500}} = 9000$$

$$(g) \text{ Energy used productivity index} = \frac{\frac{200}{(1800 \times 4)}}{\frac{100}{(500 \times 4)}} \times 100 = 55.5$$

Energy used productivity of year 2 is 55.5% of year 1.

$$(h) \text{ Raw material productivity index} = \frac{\frac{200}{(60 \times 1000)}}{\frac{100}{(10 \times 1000)}} \times 100 = 125$$

$$(i) \text{ Total factor productivity} = \frac{\text{Net Output}}{(\text{Labour} + \text{Capital}) \text{ Input}}$$

$$= \frac{\text{Total output} - (\text{Material purchased} + \text{Services hired})}{(\text{Labour} + \text{Capital}) \text{ Input}}$$

$$= \frac{200 \times 50000 - (16 \times 1000 + 15000)}{(45000 + 25000)} = \frac{153.37}{83} = 1.85$$

$$= \frac{100 \times 50000 - (10 \times 1000 + 10000)}{(40000 + 20000)}$$

The increase in total factor productivity in year 2 is 85.55 as compared to year 1.

Example 4.6

The price index for year 2006 and 2007 are given in table below. For 100 being in 1999 as base year, calculate different productivity measures for a factory whose financial statement is given below :

Sl. No.	Item	2006		2007	
		Current Rs. (in 000s)	Price Index	Current Rs. (in 000s)	Price Index
1	Net sales	1500	107	2200	130
2	Labour	200	108	380	135
3	Material	800	105	1200	140
4	Services	300	106	370	120
5	Depreciation of capital equipment and other assets (after compensated for price rise)	90	—	120	—

Solution

Given, Base Period : year 1999

1999 Price Index = 100 (assumed)

For the meaningful comparison we have to compensate the expense and receipt to the base year level. For example, net sales in year 2000 is Rs. 1500,000. Price Index in 2000 is 107 as compared to 100 in 1999. Hence, compensated sales of 2000, when brought to 1999 (base year) level would be :

$$= \frac{\text{Sales of 2000}}{\text{Price index of 2000}} \times \text{Base year Price Index}$$

$$= \frac{1500}{107} \times 100 = 1402,000$$

Similarly, other calculations are done in table below.

Sl. No.	Item	2000			2001			Productivity Rise 2000 to 2001
		(a) Current Rs. (in 000s)	(b) Price Index	Compensated at 1999 [(a) + (b)] × 100	(c) Current Rs. (in 000s)	(d) Price Index	Compensated at 1999 [(c) + (d)] × 100	
1	Net sales	1500	107	1402	2200	130	1692	
2	Labour	200	108	185	380	135	355	
3	Material	800	105	762	1200	140	857	
4	Services	300	106	283	370	120	308	
5	Depreciation	—	—	90	—	—	120	
6	Total inputs (2) + (3) + (4) + (5)			1320			1640	
7	Net output (1) - (3) - (4)			357			527	
8	Labour capital input = (2) + (5)			275			475	
9	Labour productivity (7) / (2)			1.93			1.48	- 23.32% (Fall)
10	Labour capital productivity = (7) / (8)			1.298			1.109	- 14.56% (Fall)
11	Total productivity (1)/(6)			1.602			1.032	- 2.82% (Fall)

SAQ 8

- (a) Consider the balance sheet of a company for two successive years. Analyse it from the productivity point of view.
- (b) What problems do you foresee when the revenue/expenses are not compensated for price index in Example 4.6 and SAQ 11.

Applications of Productivity Measurement

The decision making process is interlinked with the measured data. Measurements provide information and guide economic and social decisions. The quality of decision often rests on the quality of the measurements made and it is appropriate to consider measurement as a fundamental process of management. Measurement comprises quantitative data to response property of an object. The measurements help manager to decide :

- (a) What problems shall I look into?
- (b) What course of action is better? And
- (c) How well I am doing?

A proper productivity measurement system helps organisation in :

- Setting corporate goals.
- Developing plans and controls to achieve corporate objectives.
- Establishing profitability, effectiveness and efficiency of various operations.
- Evaluating returns on investment for various levels of output.
- The identification of norms to monitor productivity thus fostering productivity enhancement.
- Inter-firm comparisons.
- Developing operational (financial) budgets linked to productivity goals and targets.
- Perceiving the impact of costs and funds invested in fixed assets and inventories.

SAQ 9

In what way a productivity measurement system helps organisation?

Guidelines for Designing Productivity Measurement System

As stated earlier, the productivity measurement system should provide a measureable index which can be used to achieve the goals of productivity in an organisation. Productivity measured is directly related to the productivity improvement programme. A good productivity measurement index should have the following characteristics.

- (a) It should be meaningful, easy to understand and use and provide the status of productivity in the organisation.
- (b) It should be accurate enough to present a realistic assessment – perfect accuracy is an unreasonable expectation.
- (c) It should help in identifying the areas of low productivity so that productivity efforts can be directed to that area for improvement.
- (d) It should provide indices and information for comparison of performance in different periods.
- (e) It should provide indices and information for comparison of performance with other similar organisations/operations.
- (f) It should provide the information on inter-relationship of different sub-systems. This information can be used in allocating the resources to improve the productivity of the whole system.
- (g) It should incorporate both tangible and intangible outputs and inputs to the system.
- (h) It should consider both tangible and intangible objectives in evaluation of productivity performance.
- (i) The productivity measurement system should be hierarchical in nature, the productivity at lower levels give productivity of sub-system and productivity of sub-systems translate into overall productivity of the system.
- (j) It should facilitate to devise a reward or an incentive scheme for workers.
- (k) It should lead to the participation and involvement of employees of various levels.
- (l) It should be economical and administratively easy to run the productivity measurement system.
- (m) It should be independent from changes in monetary values and external disruptions.

Development of an Integrated Structure of Productivity Measurement

For making the productivity measurement and monitoring more effective, it should be initiated at the lowest disaggregated level, integrating the same with the performance reporting and monitoring system. For this purpose, it is necessary that a structure of productivity indicators is worked out of each centre/responsibility centre and modify the daily/weekly/monthly performance reporting system to include the aspects of productivity monitoring.

An integrated productivity measurement system includes the productivity indicators for all the production and support departments. Some of the indicators that can be incorporated at shop/department level are as follows :

Capital

Output per machine hour,
Machine down time, and
Capacity utilization.

Material

Kgs per unit output, and
Equivalent rupees per unit output.

Labour

Manpower utilisation,
Man hours per unit produced, and
Earning per hour.

Energy

KWH per hour, and
KWH per unit output.

Services

Percentage planned vs unplanned,
Maintenance hours, and
Equivalent rupees (stocks and spares) per unit output.

These indicators are then form the basis for the productivity measurement at the enterprise level.

The performance reporting system also include the information required for the enterprise level productivity measurement. The norms applicable to these indicators are required to be evolved to enable effective monitoring and control.

The typical enterprise level indicators are :

- (a) Percent capacity utilisation,
- (b) Value added per person,
- (c) Standard man hours per unit output,
- (d) Value added per rupees of wages/salaries,
- (e) Value added per rupee of capital in production assets, and
- (f) Energy (in Rs.) per unit.

SAQ 10

What should be the characteristics of good productivity measurement index?

4.10 WAYS TO IMPROVE PRODUCTIVITY

Productivity of any system can be improved by proper use of resources and by effective utilisation of systems or processes. Following action plans are required for productivity improvement :

Machine

- (a) Manual labour must be replaced by machines
- (b) Reliable machines should be used
- (c) Automation should be the key feature

Management

- (a) Management should be able to motivate workforce
- (b) There should be better planning and coordination
- (c) There should have effective control of processes

Process

- (a) There should be computerisation of systems
- (b) Use of management information systems (MIS) is recommended
- (c) Scheduling must be proper
- (d) Fast and accurate retrieval of parts is needed

Work Design

- (a) Job-design must be improved
- (b) Better work methods should be adopted
- (c) On-job training must be provided

Work Environment

- (a) Better lighting and illumination should be provided
- (b) There should be proper ventilation
- (c) There should be safety at work-place
- (d) Total Quality management (TQM) should be applied

Program

- (a) Quality circle should be implemented by the organization
- (b) Suggestion scheme must be followed
- (c) Incentive schemes should be adopted
- (d) Revise pay policy should be followed

Technology

- (a) New technology such as electro-chemical machining (ECM), etc. must be acquired
- (b) Automation in assembly for example, Surface Mounting Technology (SMT) for printed circuit board assembly unit must be acquired
- (c) Computer controlled machines such as CNC or DNC are required
- (d) Automated guided vehicle (AGV) for material transportation should be used

Manufacturing Strategy

- (a) The manufacturing system should be changed from functional to cellular layout if it is a batch production unit
- (b) Stock-less production strategy and Just-in-time framework should be adopted in the production unit
- (c) One should keep work-place clean and environment friendly (also termed as green production system)
- (d) Management should go for total change in the process/product or strategy if the system is not working properly (also known as business process re-engineering or BPR)

External Environment

- (a) There should be better political stability
- (b) There should be boosting economy and purchasing capacity of buyers.

Table 4.3 : Means to Improve Productivity for Different Planning Horizon

Planning Horizon	Means to Improve Productivity	Effect on Result
Long-term	Improve basic process by proper research and development (Use BPR) $\Rightarrow$ Heavy capital deployment	No limit to results
	Improve existing process and provide better plant and equipment utilisation (Use TQM/Kaizan) $\Rightarrow$ No major capital deployment	
Medium-term	Simplify the product, reduce and standardise the product range $\Rightarrow$ May require some capital deployment	Results are subjected to efforts and constraints
Short-term	Improve existing methods of plant operation, Improve work planning and use of manpower, Increase effectiveness of all employees $\Rightarrow$ May require little or no capital deployment	

SAQ 11

What measure would you suggest to improve productivity of a firm?

4.11 PRODUCTIVITY IMPROVEMENT TECHNIQUES IN INDUSTRIES

The achieving of maximum productivity with existing resources is a result of actions by the management, with the co-operation of workers, together with, in some cases, extra technical or scientific knowledge. These management techniques are related to product design, process design, management tools and personnel policy design.

4.11.1 Product Design

The design of the product should facilitate the use of most economical processes and methods of manufacture. This can be achieved by the close working of design and production staffs from the beginning. The productivity of a product can be undertaken at the product development stage, when production staff can examine the components and sub-assemblies and call for changes before money has been spent on production, tools and equipment. Value engineering can be applied to reduce excess work content due to design defects. The technique, systematically investigate the product and its manufacture to reduce cost and improve value.

Specification and standardisation techniques reduce the variety of products or components and helps in producing large batches on high production processes at lower overall cost.

Market, consumer and product research ensure correct quality standards. If the quality standards are higher than necessary for the efficient functioning of the product, the time taken to manufacturer it will generally be greater because of the extra care required; unnecessary rejects will also be caused. The management must be sure of the requirements of the market and of the customer, and the technical requirements of the product itself.

4.11.2 Process Design

The process planning function is responsible for specifying the machines on which the products and its components shall be made, the type of tools necessary and the speeds, feeds and other conditions under which the machines shall be run. In the chemical industries, these conditions are usually laid down by the scientists in the research departments. In all types of manufacturing industry it may be necessary to do process research in order to discover the best manufacturing techniques. Proper maintenance will ensure that plant and machinery is operating properly and will prolong its life, so reducing capital expenditure. Process planning combined with method study will ensure the selection of the most suitable tools for the operative. The layout of the factory-shop or workshop and reworking methods of the operator are designed in method study. Operators training is an aid to improving the working methods of the operator.

4.11.3 Management Tools

Management is responsible for the achievement of high productivity, especially by the reduction of ineffective time. Ineffective time can be a source of great loss even where working methods are very good. The reduction of ineffective time starts with the decision of top management concerning the markets which the firm shall try to serve. The two extreme cohesive are :

- (a) firm specialise in a small number of products make them in large quantities and sell them at lowest possible price, or
- (b) firms try to meet the special requirements of every customer.

To make many different types of product means that machines have to be stopped in order to change from one type to another; workers are unable to gain speed on work because they never have enough practice on any one job. In many organisations, variety of product grows unnoticed through attempts to make scales by meeting every special demand for variations, most of which are generally unnecessary. Specialisation is, therefore, an important step towards eliminating ineffective time.

Standardisation of components will also reduce ineffective time. By standardising most of the components in a range of models of the same type of product gives longer runs and reduces the time spent in changing over machines. Materials control aims at providing tools and materials when they are needed to reduce the idleness of workers and machines. Materials control ensures that requirements are foreseen and fulfilled in time, purchased as economically as possible and stocks are not excessive.

Breakdown of machines and plant cause idleness, reduce productivity and increase manufacturing costs. Breakdowns can be reduced by proper maintenance. Poor condition of plant and machinery produces bad products, some of which may have to be scrapped.

Good working condition is another aspect of management tools to reduce the fatigue of workers due to the effects of heat, fumes, cold or bad lighting. By proper safety arrangements worker efficiency and effectiveness can be increased by reducing the loss of time through accidents.

4.11.4 Workers Training and Personnel Policy

Personnel policy involves the whole relationship between the management and employees. If this relationship is not a good one, it is very difficult to make any management technique work satisfactorily. To create the right conditions for good relationships is part of the art of management. A sound personnel policy includes the training of managers and supervision of all ranks in proper attitudes and relations with the workers.

SAQ 12

What are the productivity improvement techniques in a manufacturing organisation?

4.12 PRODUCTIVITY IMPROVEMENT IN SERVICE INDUSTRY

Operations management initially focussed on manufacturing organizations and service operations were largely ignored. The concepts of job design, facility location, capacity, layout, inventory and scheduling can be applied to service organizations. In a service operation, the response time to customer demand is measured in minutes as compared to in manufacturing organizations, where it can be in days or weeks. In service operations, customers usually arrive at a time of their own convenience, thereby service operations may have difficulty matching capacity with demand. Still, service sector is the most important single sector of the economy due to its ever increasing contribution to GDP for many countries. The objective of service sector is primarily to provide infrastructural support to manufacturing sector and to bring about improvement in quality of life by fulfilling the diversifying service needs for higher standard of living.

In a typical situation, the cost of an unhappy customer can be calculated as :

- (a) How many complaints have you received in the past month?

This is called the BASE RATE.

- (b) For every complaint received, the average company has, in fact 6 other customers with problems.

Multiply the Base Rate by 6. This is called the TRUE BASE RATE. This is the number of people who have experienced a customer problem and have not complained to you. The complainers are more likely to do business with you than the non-complainers.

- (c) 90% of those customers who complain will not come back without positive response. Let us assume that only 25% of the seriously dissatisfied customers opt to go else where next time.

- (i) Divide True Base Rate by 4.

- (ii) Multiply this by the value of the customer.

This value is the COST OF CUSTOMER DISSATISFACTION.

- (d) It costs about 5 times more to secure a new customer than it does to retain an existing one. Multiply the cost of customer dissatisfaction by 5.

This is the cost of losing customers on a yearly basis due to poor customer service.

- (e) The average customer with a complaint will tell 9 to 10 people about that complaint. In fact 13% of the customer with a complaint tell more than 20 people.

Multiply the True Base Rate by 10.

This is the NEGATIVE IMAGE RATE. This is the number of people who heard bad things about your company and may even decide not to do business with your company.

[This is based on a study in 1985 in USA.]

The main difficulty in productivity of service sector is in measuring output. Consider for example, health services. If a doctor's first concern is to safeguard the patient health, how one can measure doctor's output. If hospital administrator's concerns is to provide the facilities that may be needed by a doctor and the services required for patients. In this case also it is difficult to define the measures of productivity.

The two principal paths for productivity improvement in the service sector appear to be via investments in equipment and more efficient personnel practices. The latter is prominent because services are highly labour intensive.

Investments in Equipment

The introduction of new technology to improve performance is a logical, non-controversial way to increase output of the proper technology is identified and available. Computer based systems have had a major input in service areas. Computerised reservation system of Indian Railways have increased the passenger handling capacity of the reservation counters. For better results it is important to identify the right technology and then the perfect implementation.

Personnel Practices

More efficient employees activities at all levels is greatest potential area in service sector. It is very difficult to measure the output of directors, clerks, teachers, etc. and therefore it is hard to set standards by which performance can be evaluated. The crude feedback is inadequate for control of operations. Employees accountability can be developed by measuring how well jobs are done so as to improve future progress. Different approaches can be :

- (a) improving communication between supervisors and subordinates;
- (b) motivating employees;
- (c) establishing a data base for productivity improvement; and
- (d) better administrative decision on wages, production, etc.

SAQ 13

Identify the techniques help in improving productivity in service sector.

4.13 SUMMARY

Now the difference between production and productivity is clearly explained in this unit. Production is output or performance measures in absolute terms with no relationships with resources spent. Productivity essentially represents the relationship between the performance and input. However production, productivity and quality cannot be considered in isolation. Both production and productivity have a common dimension in the form of output.

In this unit you have studied about productivity and its effect on society. Productivity is a quantitative measure of performance. It is different from production which is activity of producing goods or services. Productivity is related to the efficient utilisation of input resources into produced output in the form of value added goods or services.

Productivity of any system can be improved by either proper use of resources or by effective utilisation of system or process. Increase in production does not necessarily mean increase in productivity. Productivity is always associated with the context in which it is calculated. For example, labour productivity, material productivity, capital productivity and energy productivity. Productivity at National level is discussed to give you an idea of importance of productivity for the economic growth of country.

Productivity is the relationship between the output generated by a production or service system and the input provided to create this output. For any productivity improvement drive, it is important to identify the major factors, the sub-factors and sub-sub factors. It is not only the internal factors which are under the control of management, the external factors also should be identified and there by the policies of the organisation must change to improve productivity in the long run.

In this unit you have studied about various factors which affect productivity at organisational level and national level. At the organisational level, the factors are classified as external factors and internal factors. The external factors are those which are beyond the control of the industrial organisation and the internal factors are those which are within its control. At the national level, the capital productivity and the material productivity are the two major areas where improvements can lead to a substantial increase in national productivity.

The concept of productivity is also increasingly linked with quality and employment. An element of key importance is the quality of the workforce, its management and its working conditions. It has been generally recognised that rising productivity and improving quality of working life and employment do tend to enhance one-another.

After study this unit, you now able to measure productivity of a manufacturing unit. For this purpose, no suitable yard stick has been devised so far. One can compare output with man-hour input, or with the total of labour and capital input. In this unit productivity has been defined from the point of view of economists, engineer, accountant, psychologist and manager, because productivity importance varies with its function. If you observe the various definition you will find that partial productivity, total factor productivity and total productivity are the three types of productivity measures. Partial productivity includes labour productivity, material productivity, capital productivity and energy productivity. Total factor productivity is ratio of net output to (labour + capital) input. Total productivity is ratio of total output (product + service) to the total input (labour + material + capital + energy + other expenses). Favourable, satisfactory and unfavourable productivity trends explained in this unit will be helpful to you in understanding the action required for improvement. As a manager/engineer you should be able to explain the consequence of such trends to your top management.

4.14 KEY WORDS

Production	: Production is a sequence of operations that transforms input of given form into desired form of output with the help of man, material, machine, money and equipment.
Productivity	: Productivity signifies how well a system is performing against the objectives or goals. Productivity essentially represents the relationship between the performance (or output) and input.
Partial Productivity	: It is defined on the basis of the class of the input being considered.
Total Factor Productivity	: It is the ratio of net output to the sum of labour and capital inputs.
Total Productivity	: It is the ratio of total output to the sum of all input factors.
Partial Productivity	: Labour Productivity, Material Productivity, Capital Productivity and Energy Productivity are the examples of partial productivity.
Human Development Index (HDI)	: Adopted by UNIDO is based on the level of consumption as a direct measure of the economic development and rate of its process.
External Factors of Productivity :	The external factors are those which are beyond the control of the individual organisation. By proper understanding of these factors, actions can be taken to improve the productivity in long run.

- Internal Factors of Productivity :** The internal factors are within the control of the organisation like product, plant and equipment, people, and work methods.
- Hard Factor :** Factors that cannot be changed easily and require stronger financial and organisational interventions.
- Soft Factor :** Easily changeable factors like people, organisation and systems, work methods, and management styles.
- Partial Productivity :** Partial productivity is the ratio of the output to one class of input. Labour, capital and material productivity are the examples of partial productivity.
- Total Factor Productivity (TFP) :** Total factor productivity is the ratio of the net output to the sum of associated labour and capital (factor) input.
- Total Productivity :** Total productivity is the ratio of total output to the sum of all input factors.
- Multifactor Productivity :** Multifactor productivity (MFP) index shows the rate of change in "productive efficiency", and is obtained as the ratio of output to a weighted combination of labour and capital inputs.
- Unit Labour Cost Index (ULC) :** Unit labour cost is the remuneration of labour to produce one unit of output. It is computed as the ratio of the labour cost index to an index of production.
- Hourly Labour Cost :** Hourly labour cost is the ratio of compensation to total hours worked, inclusive of overtime.
- Gross Domestic Product per Capita :** The Gross Domestic Product (GDP) gives the total value of goods and services produced in the country and the GDP per capita is commonly used as an indicator for the standard of living of the population.

4.15 ANSWERS TO SAQs

Refer the preceding text for all the Answers to SAQs.